QUASI-PROJECTIVITY DESCENDS ALONG AN ARBITRARY FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the quasi-projectivity conclusion; no step is referred to the reader.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-quasi-projective-after-field-extension is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Finiteness of generation descends along a field extension. Algebra, Lemma algebra-lemma-finite-type-descends, invoked at proof lines 2830-2831. Supplies that the coordinate rings of affine opens of X are finitely generated k-algebras.
- Ampleness descends along a field extension. Same chapter, Lemma lemma-ample-after-field-extension, invoked at proof lines 2838-2839. Supplies the ample invertible sheaf on X.
- Quasi-projectivity is equivalent to local finite type, quasi-compactness and an ample invertible sheaf. Morphisms, Lemma morphisms-lemma-finite-type-over-affine-ample-very-ample, invoked at proof lines 2835-2837 and again at 2840-2841. Supplies the unpacking of the definition used at the start and the closing step.

Source reference. Original lemma 00QP: sources/stacks-chow/tags.txt, source lines 975–975. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BDC: sources/stacks-varieties/pinned-original-source.tex, source lines 2740–2740. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01VT: sources/stacks-chow/tags.txt, source lines 2458–2458. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. If X_K is quasi-projective over K for some field extension K/k, then X is quasi-projective over k.*

Proof. By definition a morphism of schemes $g : Y \to T$ is quasi-projective if it is locally of finite type, quasi-compact, and there exists a g-ample invertible sheaf on Y. Let K/k be a field extension such that X_K is quasi-projective over K. Let $\mathrm{Spec}(A) \subset X$ be an affine open. Then U_K is an affine open subscheme of X_K, hence A_K is a K-algebra of finite type. Then A is a k-algebra of finite type by Algebra, Lemma 00QP. Hence $X \to \mathrm{Spec}(k)$ is locally of finite type. Since $X_K \to \mathrm{Spec}(K)$ is quasi-compact, we see that X_K is quasi-compact, hence X is quasi-compact, hence $X \to \mathrm{Spec}(k)$ is of finite type. By Morphisms, Lemma 01VT we see that X_K has an ample invertible sheaf. Then X has an ample invertible sheaf by Lemma 0BDC. Hence $X \to \mathrm{Spec}(k)$ is quasi-projective by Morphisms, Lemma 01VT. □